WASP-135b

R-0081

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-135_250528` · created 2026-07-15T21:42:08+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460823.87 +/- 0.014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.075 +/- 0.046
Transit depth (Rp/Rs)^2	0.56 +/- 0.69 [%]
Orbital Inclination (inc)	84.83 +/- 2.85
Airmass coefficient 1 (a1)	0.951 +/- 0.013
Airmass coefficient 2 (a2)	0.048 +/- 0.065
Scatter in the residuals of the lightcurve fit is	1.38 %
Best Comparison Star	#2 - [195, 285]
Optimal Aperture	2.68
Optimal Annulus	9.34
Transit Duration (day)	0.015 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.075 ± 0.046 vs 0.139 (deviation 1.39σ)
Duration fit vs published	0.015 d vs 0.0687 d (deviation 4.13σ)
Mid-time O–C	-33.3 min
Depth SNR	1.6
Residual scatter	1.38 %

Light curve

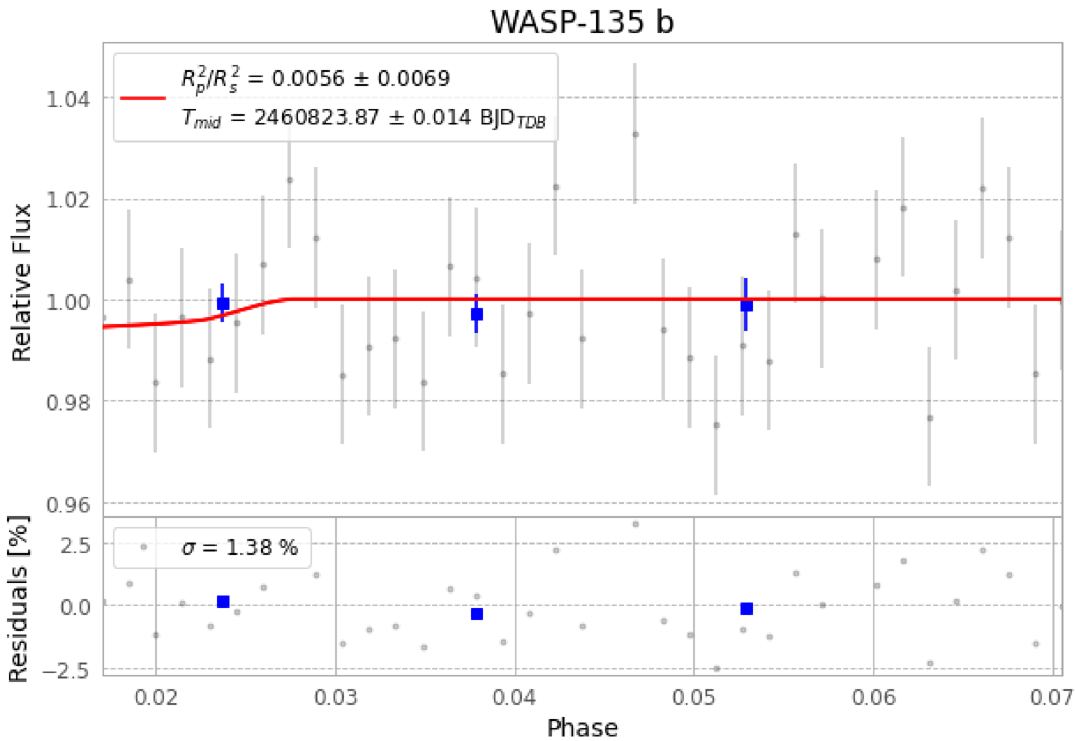

Figure 1 — FinalLightCurve_WASP-135 b_2025-05-28.png (SHA-256 425d890b4c91e1ce…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [351, 285], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 7b72733ff96fc799…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13`

Data provenance

37 FITS frames (24 MB), `WASP-135250528092115.FITS` → `WASP-135250528110915.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-135-250528.log (56a2aa03a103…), FinalLightCurve_WASP-135 b_2025-05-28.png (425d890b4c91…), FinalLightCurve_WASP-135 b_2025-05-28.csv (070d11e317b5…)

Compute resources

Wall time 105.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)